Classification of States

Definition. Accessibility

State j is **accessible** from state i, denoted by $i \to j$, if

$$p_{ij}^{(n)} > 0 \quad \text{for some } n \geq 0.$$

Remark.

- Accessibility is not a one-step concept: p_{ij} may be zero while $p_{ij}^{(n)} > 0$ for some $n > 1$.
- It is not symmetric: $i \to j$ does not imply $j \to i$.
- Since $p_{ii}^{(0)} = 1$, every state is accessible from itself.

Definition. Communication

States i and j **communicate**, denoted by $i \leftrightarrow j$, if

$$i \to j \qquad \text{and} \qquad j \to i.$$

Note. Communication is an **equivalence** relation.

Definition. Communication Class

A **communication class** is a maximal set $\mathscr{C} \subseteq S$ whose states communicate with one another.

Note. Communication classes form a partition of the state space S.

Definition. Irreducible chains

A Markov chain is **irreducible** if it has exactly one communication class. Otherwise, it is **reducible**.

Definition. Closed Class

A communication class $\mathscr{C}$ is **closed** if the chain cannot leave it:

$$p_{ij} = 0 \qquad \text{for every } i \in \mathscr{C} \text{ and } j \notin \mathscr{C}.$$

Remark.

- Once $X_n \in \mathscr{C}$ for any n, all future states X_{n+m} remain in $\mathscr{C}$ with probability 1, any $m \geq 0$.
- An **absorbing state** i satisfies $p_{ii} = 1$. Hence $\{i\}$ is a closed class.
- A closed class may contain several states and the chain may continue moving inside it.

Theorem

Every finite-state Markov chain has at least one closed communicating class.

Definition. Hitting and First-Return Times

For a state j, define

$$T_j = \inf\{n \geq 0 : X_n = j\}.$$

For a process **starting from state** i, define the **first *return* time**

$$T_i^+ = \inf\{n \geq 1 : X_n = i\}.$$

Remark.

- T_j: the first time to arrive at state j.
- T_i^+: we know that we start from i, and we wonder the first return time to i.

Definition. First-Return Probability

Starting from i, the probability that the first return occurs exactly at time n is denoted as

$$f_{ii}^{(n)} = P_i(T_i^+ = n), \qquad n \geq 1.$$

The probability of ever returning is

$$f_{ii} = P_i(T_i^+ < \infty) = \sum_{n=1}^{\infty} f_{ii}^{(n)}.$$

Takeaway. Consider state i. If the process returns to i almost surely (with prob. 1), i.e. $f_{ii} = 1$, then it will return to i infinitely many times, and the total returning prob. $\sum_{n=1}^{\infty} p_{ii}^{(n)}$ is infinite. We call i **recurrent**. This is a class property.

Definition. Recurrence and Transience

A state i is **recurrent** if

$$P_i(T_i^+ < \infty) = 1.$$

It is **transient** if

$$P_i(T_i^+ < \infty) < 1.$$

Define the number of visits to i after time zero:

$$N_i = \sum_{n=1}^{\infty} \mathbf{1}\{X_n = i\}.$$

Theorem. Number of Revisits

For every $m \geq 1$,

$$P_i(N_i \geq m) = (f_{ii})^m.$$

Consequently,

$$\begin{cases} N_i = \infty \text{ a.s.}, & f_{ii} = 1, \\ N_i < \infty \text{ a.s.}, & f_{ii} < 1. \end{cases}$$

$$E_i[N_i] = \begin{cases} \dfrac{f_{ii}}{1 - f_{ii}}, & f_{ii} < 1, \\ \infty, & f_{ii} = 1. \end{cases}$$

Hence,

$$i \text{ recurrent} \iff N_i = \infty \text{ a.s.} \iff E_i[N_i] = \infty,$$

$$i \text{ transient} \iff N_i < \infty \text{ a.s.} \iff E_i[N_i] = \frac{f_{ii}}{1 - f_{ii}}.$$

Define the n-step return probability

$$p_{ii}^{(n)} = P_i(X_n = i).$$

Theorem. Return-Probability Criterion

$$i \text{ is recurrent} \iff \sum_{n=1}^{\infty} p_{ii}^{(n)} = \infty,$$

$$i \text{ is transient} \iff \sum_{n=1}^{\infty} p_{ii}^{(n)} < \infty.$$

Starting from state i,

$$\begin{aligned} i \text{ recurrent} &\iff P_i(T_i^+ < \infty) = f_{ii} = 1 \\ &\iff P_i(N_i = \infty) = 1 \\ &\iff E_i[N_i] = \infty \\ &\iff \sum_{n=1}^{\infty} p_{ii}^{(n)} = \infty. \end{aligned}$$

Remark.

- $f_{ii}^{(n)} = P_i(T_i^+ = n)$ counts paths whose *first* return occurs at time n.
- $p_{ii}^{(n)} = P_i(X_n = i)$ counts all paths that are at i at time n, including paths that returned earlier.

Note. So, $f_{ii}^{(n)} \leq p_{ii}^{(n)}$.

Theorem

If $i \leftrightarrow j$, then i and j are either both recurrent or both transient.

Theorem

For a **finite-state** Markov chain:

- at least **one closed communicating class** exists;
- recurrent class $\iff$ closed class;
- all states in a nonclosed class are transient.

Definition. Period

For a state i, define

$$d(i) = \gcd\{n \geq 1 : p_{ii}^{(n)} > 0\}.$$

If the set is empty, use the convention $d(i) = 0$.
A state is **aperiodic** if $d(i) = 1$.

Theorem

If $i \leftrightarrow j$, then

$$d(i) = d(j).$$

Definition. Aperiodic Chain

A Markov chain is **aperiodic** if every state is aperiodic, i.e. $d(i) = 1$ for all $i \in S$.

Theorem

Let state i have period $d = d(i) \geq 1$.

- For every $n \geq 1$,

$$d \nmid n \quad \Longrightarrow \quad p_{ii}^{(n)} = 0.$$

- There exists an integer $N_i \geq 1$ such that

$$p_{ii}^{(kd)} > 0 \qquad \text{for every integer } k \geq N_i.$$

Note. Some small multiples of d may not be possible return times.

Definition. Regularity

A **finite-state** Markov chain with transition matrix $\mathbf{P}$ is **regular** if there exists an integer $r \geq 1$ such that **every entry of $\mathbf{P}^r$ is strictly positive**; that is,

$$p_{ij}^{(r)} > 0 \qquad \text{for all } i, j \in S.$$

Theorem

For a finite-state Markov chain,

regular $\iff$ irreducible and aperiodic.

Theorem

Let $\{X_n\}$ be a **regular** Markov chain on a finite state space S, with transition matrix $\mathbf{P}$. There exists a distribution $\boldsymbol{\pi} = (\pi(j))_{j\in S}$ such that, for every initial distribution $\boldsymbol{\pi}_0$,

$$\lim_{n\to\infty} \boldsymbol{\pi}_n = \lim_{n\to\infty} \boldsymbol{\pi}_0 \mathbf{P}^n = \boldsymbol{\pi}.$$

- The convergence is entrywise convergence, i.e. $\pi_n(j) \to \pi(j)$ for all $j \in S$.
- This limit is independent of $\boldsymbol{\pi}_0$, i.e. $p_{ij}^{(n)} \to \pi(j)$ for all $i, j \in S$.
- $\boldsymbol{\pi}$ is unique, and it satisfies

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}, \qquad \sum_{j\in S} \pi(j) = 1, \qquad \pi(j) > 0.$$

To calculate the limiting distribution, solve

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}, \qquad \sum_{j\in S} \pi(j) = 1, \qquad \pi(j) \geq 0.$$

- Before calculation, **verify the convergence first!**
- This is a **linear system** in the unknowns $(\pi(j))_{j\in S}$.

Remark. The jth entry $\pi(j)$ of $\boldsymbol{\pi}$ can be interpreted as the **long-run fraction of time spent in state** j:

$$\left(\frac{1}{n}\sum_{k=1}^{n} \mathbf{1}_{\{X_k = j\}}\right)_{j\in S} \xrightarrow{\text{a.s.}} \boldsymbol{\pi}.$$

This convergence holds with probability 1.

Definition. Positive and Null Recurrence

For a recurrent state j, let $m_j = \mathbb{E}_j[T_j^+]$, where T_j^+ is the first return time to state j.

- **Positive** recurrent: $m_j < \infty$.
- **Null** recurrent: $m_j = \infty$.

Remark.

- Recurrence guarantees a return, but the mean may be infinite. Positive recurrence further guarantees a finite return time.
- Positive and null recurrence are class properties.
- Every finite irreducible chain is positive recurrent.

Definition. Ergodicity

A Markov chain is **ergodic** if it is irreducible, **positive recurrent**, and aperiodic.

Theorem. Limiting Distribution

An ergodic chain on a finite or countable state space S has a unique limiting distribution $\boldsymbol{\pi} = (\pi(j))_{j\in S}$. For every initial distribution $\boldsymbol{\pi}_0$,

$$\lim_{n\to\infty} \pi_n(j) = \pi(j) = \frac{1}{m_j} > 0, \qquad j \in S.$$

- The limit is independent of $\boldsymbol{\pi}_0$.
- It satisfies

$$\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}, \qquad \sum_{j\in S} \pi(j) = 1.$$

- For finite chains, ergodicity is equivalent to regularity.
- Irreducible + aperiodic + finite states ⇒ limiting dist.
- Irreducible + aperiodic + positive recurrent ⇒ limiting dist.